

Aditya Pol

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Education

Pimpri Chinchwad College of Engineering <i>B.Tech – Computer Science Engineering (AI & ML)</i> CGPA: 7.8	Pune, India 2023 – 2027
Ligade Patil Jr College of Science <i>12th Grade</i> Percentage: 80.67% — MHT-CET: 97.93 %ile	Karad, Satara 2023
Yashwant High School <i>10th Grade</i> Percentage: 97.20%	Karad, Satara 2021

Projects

AI Resume Analyzer & Job Preparation Platform (LLMs, RAG, Node.js, React)

Full Stack + Generative AI Project

- Built an AI-powered platform that analyzes resumes and provides personalized job preparation insights using LLM APIs.
- Implemented Retrieval-Augmented Generation (RAG) using FAISS/ChromaDB for context-aware recommendations.
- Designed prompt pipelines for resume feedback, skill gap analysis, and interview question generation.
- Integrated backend APIs with React frontend to deliver real-time AI responses.

Customer Review Sentiment Analysis and Topic Modeling (Python, NLP, Scikit-learn)

Machine Learning Project

- Built an end-to-end NLP pipeline processing 1000+ customer reviews.
- Performed preprocessing including tokenization, stemming, and feature extraction (TF-IDF).
- Trained Logistic Regression model for sentiment classification and evaluated using F1-score.
- Applied topic modeling using Truncated SVD to extract key themes from reviews.

3-Layer Deepfake Detection Pipeline (CRAVE System)

AI Security + Deep Learning Project

- Designed a multi-modal deepfake detection system combining video, audio, and behavioral signals.
- Extracted facial landmarks using MediaPipe FaceMesh and computed visual artifact scores.
- Processed audio using MFCC features to detect synthetic speech patterns.
- Built an MLP-based fusion model combining lip-sync, motion, and prosody signals for final classification.

Technical Skills

- **Programming:** C++, Python, Java
- **Generative AI:** LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents
- **AI / ML:** Machine Learning, Deep Learning, NLP, Model Evaluation
- **LLM Frameworks:** LangChain, LlamaIndex, Hugging Face Transformers
- **Vector Databases:** FAISS, Pinecone, ChromaDB
- **Web Development:** Node.js, Express.js, REST APIs
- **Databases:** MySQL, MongoDB
- **Cybersecurity:** Deepfake Detection, Multimedia Forensics, AI Security
- **Core CS:** Data Structures & Algorithms, OOP, Computer Networks
- **Tools:** Git, GitHub, Docker, VS Code, Jupyter Notebook

Achievements

- Secured **2nd Rank** in the **Secure AI Software & Systems Hackathon** at **BITS Pilani, Goa** in the **Deepfake / Voice Detection** problem statement.

Work Experience

AI Research Intern – PCCOE, Pune

May – July 2025

- Conducted research on Quantum AI in healthcare by analyzing 20+ research papers.

GRE/GMAT Coordinator – Student Development Welfare (SDW), PCCOE

- Lead GRE/GMAT awareness initiatives and coordinate mentoring programs.